BOND PRICING

FINANCIAL ECONOMICS – SPRING 2026

DEFINITIONS

- A bond is issued by a government (sovereign bond) or a company (corporate bond) and is characterized by three main elements:
  - the face value of the bond
  - maturity, i.e. the date on which the issuer of the bond promises to repay the nominal value
  - the coupon, i.e. a remuneration that is paid periodically (every six months or every year) by the issuer of the bond to the holder of the bond. The coupon can be fixed in monetary value or vary over time, for example depending on the trend in short-term interest rates. The coupon is equal to the product of the coupon rate and the nominal value. Securities on which a coupon is not paid are defined as zero-coupon bonds

EXAMPLE

- Example, a bond with a nominal value of € 1,000 and maturity 31 December 2025 with a coupon rate of 2% will pay the holder a coupon of € 20 each year until the end of 2025 and on 31 December 2025 will repay € 1,000 to the holder of the bond

The coupon rate is expressed as an annual value. Some bonds may pay the coupon on a semi-annual basis, in this case paying the annual coupon divided by 2

REAL BONDS

- Real bonds pay a fixed coupon rate on a face value that rises every year with the rate of inflation

In this way, the annual coupon is fixed at real value rather than at face value, and the same happens at the final value

Period	Inflation	Nominal value	Coupon	Capital	Total
0		1000,00			
1	2%	1020,00	40,80		40,80
2	3%	1050,60	42,02		42,02
3	1%	1061,11	42,44	1061,11	1103,55

CORPORATE BONDS

- They are issued by companies. The details of the issue are described in a regulation (indenture) that describes elements such as collateral, dividend policy, additional funding, presence of call or put options
- Corporate bonds are rated by one or more of the three major agencies Standard & Poor's, Moody's, and Fitch. The rating is assigned considering the financial health of the issuer and therefore the probability of repayment of interest and principal.
- Investment grade bonds generally have lower risk (but Lehman Brothers had issued investment grade bonds) and therefore pay less interest
- Non-investment grade bonds are considered "speculative investments" and pay higher interest to compensate for the higher risk

The change in the issuer's health over time may result in a change in the rating

SECONDARY MARKET PRICING

- Bonds are traded on secondary markets where the market price is determined, which is generally different from the face value
- The prices observed on secondary markets do not represent the amount of money actually paid by the buyer, since the seller must be remunerated for the value corresponding to the current coupon
- “Flat price" does not take into account the current coupon. The “invoice" price is equal to the sum of the dry price and the interest accrued from the last coupon until the day on which the purchase and sale transaction is settled ("accrual")

SECONDARY MARKET PRICING

- A bond maturing on 31/12/2025 has a flat price on the secondary market of 98 on 31/3/2019.
- The coupon of € 20 matures proportionally over time, and therefore that at the end of March a value of € 5 is attributable to the person holding the bond at that time (accrual).
- The buyer of the bond at the end of March must therefore pay an invoice price of € 98 + € 5 = €103
- There are different conventions for calculating the accrual, for example 30/360 indicates that each of the 12 months of the year is considered composed of 30 days and the length of the year is 360 days

PRICING

- To determine the secondary market price of a bond, we apply the definition of the present value of future cash flows, i.e. coupons and face value, at the moment assuming that there is only one interest rate in the market:

$$P = \sum_{t=1}^{T} \frac{C}{(1+i)^t} + \frac{FV}{(1+i)^T}$$

- C = 0 is the zero-coupon bond
- Some bonds may pay the coupon semi-annually. In this case, the evaluation takes place considering $C/2$ in the numerator and $i/2$ in the denominator and of course doubling the number of terms to 2×T

PRICING

- The price can also be found from:

$$P = C \times \frac{1}{i}\left[1 - \frac{1}{(1+i)^T}\right] + FV \times \frac{1}{(1+i)^T}$$

- The first term is the "Annuity factor", the present value of a constant payment for T years discounted at the rate I

- The second term is the "Face Value factor", the present value of the final payment

The bond pricing formula highlights the existence of an inverse relationship between the interest rate used in the denominator and the price. Rising market interest rate at which coupons are discounted causes the bond's secondary market price to fall

PRICING

- The above formula can be rewritten as follows:

$$P = \mathrm{c} \times FV \times \frac{1}{i}\left[1 - \frac{1}{(1+i)^T}\right] + FV \times \frac{1}{(1+i)^T} =$$

$$= \mathrm{FV}\left\{\mathrm{c} \times \frac{1}{i}\left[1 - \frac{1}{(1+i)^T}\right] + \frac{1}{(1+i)^T}\right\}$$

Showing that if $\boldsymbol{c} = \boldsymbol{i}$ the price of the bond is equal to the face value

PRICING: INFINITE LIFE

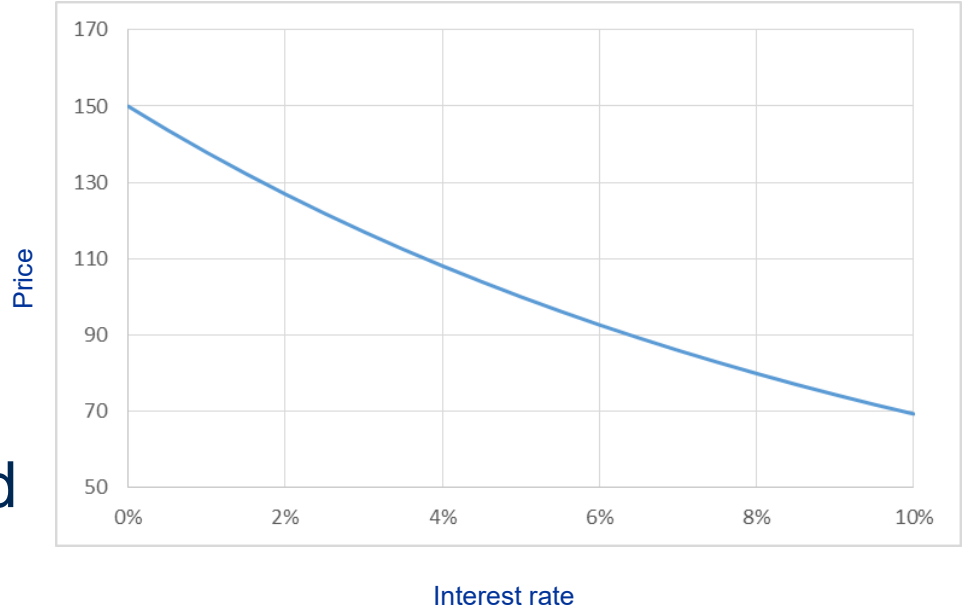
- An irredeemable security is a bond with an infinite life, The pricing formula for such a security can be written as follows:

$$P_{t,\infty} = \frac{C}{1+i} + \frac{C}{(1+i)^2} + \cdots = \frac{C}{i}$$

therefore the price of an irredeemable security is equal to the ratio between coupon and rate

PRICING: PICTURES

- Why does the secondary market price move in inverse proportion to the interest rate? Let's assume that new bonds are issued at any time by paying a coupon rate equal to the value of the market interest rate. When the interest rate subsequently rises (falls) investors will pay less (more) to be able to buy a previously issued bond that pays a lower (higher) coupon rate
- The function that links the price to the interest rate is convex: an increase in the rate implies a reduction in the price in absolute value that is less than the increase in the price resulting from an equal reduction in the rate



EXAMPLES

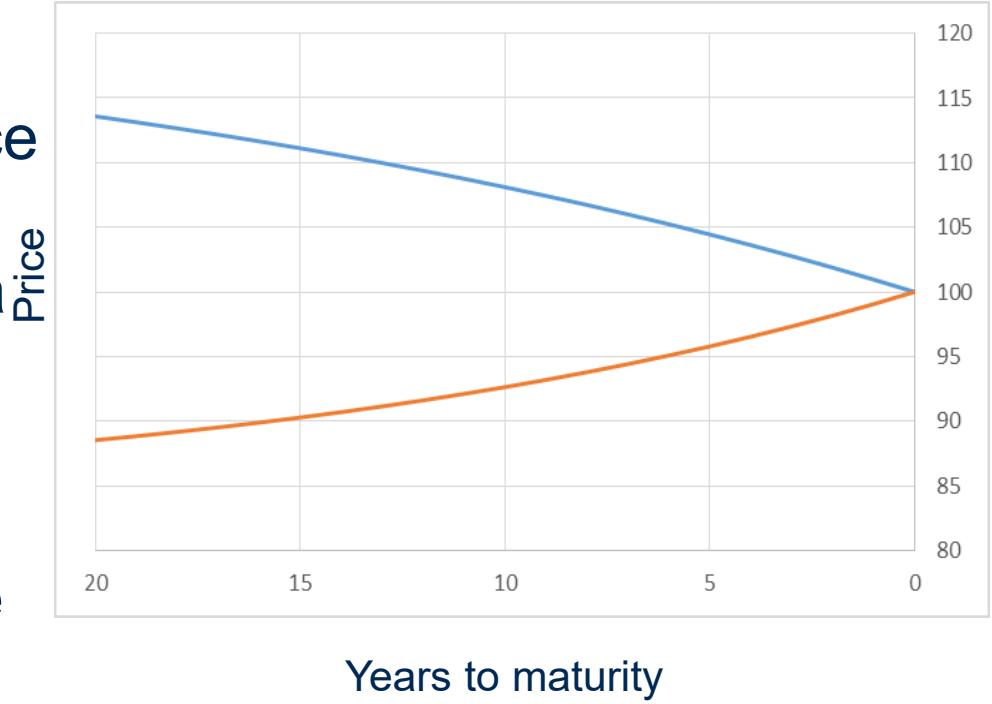
- The longer the maturity of the bond, the greater the sensitivity of the price to changes in the interest rate

- The prices in the matrix are determined based on the formula $P = C \times \frac{1}{i}\left[1 - \frac{1}{(1+i)^T}\right] + FV \times \frac{1}{(1+i)^T}$

	tasso				
Vita residua	2%	4%	6%	8%	10%
1 anno	1059,11	1038,83	1019,13	1000,00	981,41
10 anni	1541,37	1327,03	1148,77	1000,01	875,35
20 anni	1985,04	1547,11	1231,15	1000,02	828,41
30 anni	2348,65	1695,22	1276,76	1000,03	810,71

PRICING: PICTURES

- At the same rate, the price converges over time to the nominal value of the bond (price drift)
- The picture describes the two cases of a "premium" bond, in which the market price is initially higher than the nominal value, and a "discounted" bond, in which the market price is initially lower than the nominal value. When the market price of the bond is equal to the face value, the bond is "at par“



- Over time, and all things being equal, both tend to converge towards the nominal value

In fact, as time passes, the number of coupons to be discounted to determine the price decreases. The decreasing price of the premium bond is offset by the high coupon rate, and vice versa for the discount bond

PRICING: INFINITE LIFE

- When the bond is held for a period shorter than its maturity, it becomes important to describe its periodic yield, which includes the coupon paid and the percentage change in the price of the security

$$h(t, t+1, T) = \frac{\mathrm{C} + P_{t+1,T} - P_{t,T}}{P_{t,T}}$$

- It is a random variable that depends on the price trend of the security that. As we will see in the next chapter, it fluctuates based on many elements including the remaining life of the bond

YIELD TO MATURITY

- Let's assume that we observe the price and look for a synthetic indicator of the "average profitability" of the bond if held to maturity

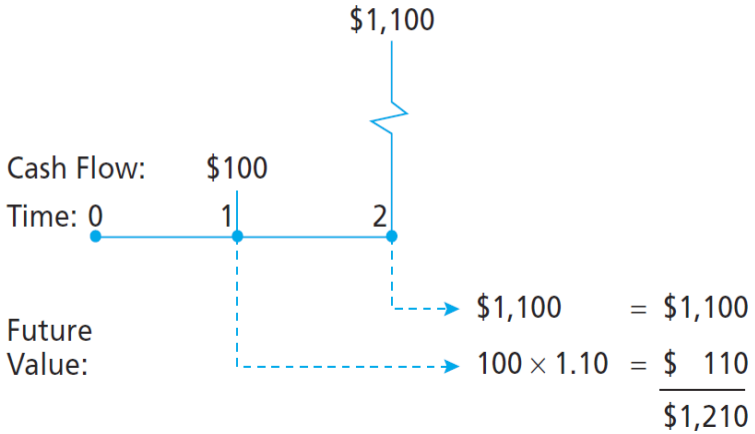
Yield to maturity (YTM) is the rate used to discount future cash flows that makes the present value of coupons and face value equal to the price of the bond. To calculate the YTM, the following equation must therefore be solved for the unknown y:

$$P = \sum_{t=1}^{T} \frac{C}{(1+y)^t} + \frac{FV}{(1+y)^T}$$

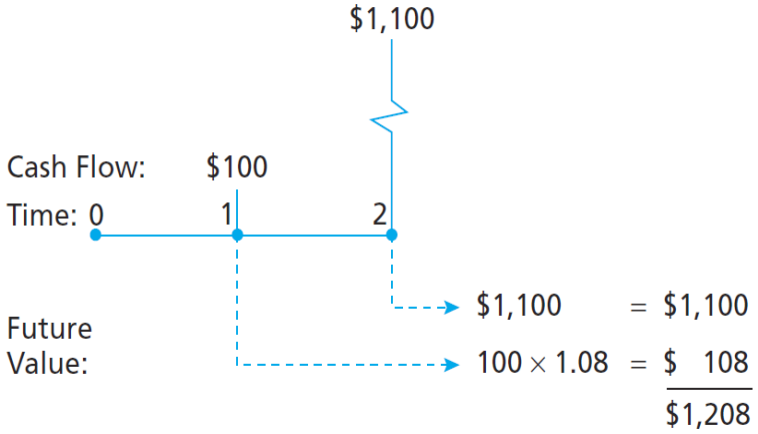
- The assumption is that all coupons are reinvested at the same YTM. If the actual reinvestment takes place at rates lower than the YTM, then the YTM realized ex post is different from the one calculated ex ante

YIELD TO MATURITY AND REINVESTMENT

A: Reinvestment Rate = 10%



B: Reinvestment Rate = 8%



DEFAULT YIELD TO MATURITY

- The market rate and the price of the bond are determined by investors on the basis of various elements, including maturity (see next chapter on the term structure) that introduces the structure of rates by maturity and investors' perception of risk in relation to the issuer's ability to pay coupons and principal
- In case of default, creditors obtain only a share of their claim, which depends on the value of the company's assets at the time of default

DEFAULT YIELD TO MATURITY

- Investors estimate how likely default is likely to occur (probability of default, PD) and what the loss could be (in terms of the present value of the coupons that have yet to be paid and the payment of the nominal value) in the event of default (loss given default, LGD)
- The higher PDs and LGDs, the lower the price investors are willing to pay to buy the bond and therefore the higher the interest rate. A higher interest rate compensates investors for the risk they take
- It is often assumed that only bonds issued by companies (corporate bonds) are subject to the risk of default. In fact, investors also value PD and LGD for sovereign bonds
- Rating companies issue judgments based on proprietary models that help investors estimate PDs and LGDs

EXAMPLE

- One-year corporate ZCB. If the risk-free rate is 5%, the market price is

$$P = \frac{100}{1.05} = 95.23$$

- If PD is 10% and LGD is 50% of the nominal value, the price, in the event of a risk-neutral market, will be equal to:

$$P = \frac{0.9 \times 100 + 0.1 \times 0.5 \times 100}{1.05} = 90.47$$

- Investors are compensated for the Expected Loss by means of a higher YTM. If investors were risk-averse, the market price would be below 90.47 and the YTM even higher